

Banking Database Analysis Project

SQL Portfolio Project

Name: Lerato Lehoko

Bachelor of Science (Biochemistry & Statistics)

Aspiring Data Analyst

Date: July 2026

Table of Contents

COVER PAGE	1
Introduction.....	3
2. Project Objectives	3
3. Database Design	4
4. Database Tables.....	4
5. Business Questions and Analysis	5
6. SQL Skills Demonstrated	14
7. Key Findings	15
Customer Analysis.....	15
Branch Performance	15
Transaction Analysis	16
Loan Analysis	16
Customer Distribution.....	16
8. Recommendations	16
9. Conclusion	18

Introduction

The Banking Database Analysis Project was developed to demonstrate practical SQL skills through the design and analysis of a relational banking database. The project simulates a real-world banking environment by modelling customers, bank branches, employees, customer accounts, financial transactions and loans.

The database was designed using MySQL and populated with realistic sample data to replicate common banking operations. SQL queries were then written to answer business questions that support decision-making, customer analysis, branch performance evaluation and financial reporting.

This project demonstrates the ability to design databases, write analytical SQL queries and transform raw data into meaningful business insights. The techniques used throughout the project reflect practical skills required for entry-level data analyst positions.

2. Project Objectives

The objectives of this project were to:

- Design a relational banking database using MySQL.
- Establish relationships between multiple tables using primary and foreign keys.
- Populate the database with realistic banking data.
- Analyze banking information using SQL.
- Apply SQL techniques to answer real-world business questions.
- Demonstrate data analysis skills suitable for a professional portfolio.
- Present analytical findings and recommendations based on the results.

3. Database Design

The banking database consists of six related tables designed to simulate the core operations of a commercial bank. Each table stores specific business information while maintaining relationships through primary and foreign keys.

The design follows the principles of relational database management by minimizing data redundancy and maintaining data integrity.

The relationships between the tables allow customer information, accounts, transactions and loans to be analyzed together to generate meaningful business insights.

4. Database Tables

Table	Description
Customers	Stores customer personal information including names, gender, province, city and joining date.
Branches	Stores information about the bank's South African branches.
Employees	Stores employee details and the branch where each employee works.
Accounts	Stores customer account information including account type, balance and status.
Transactions	Stores deposits and withdrawals performed on customer accounts.

Loans	Stores loan information including loan type, loan amount, interest rate and loan status.
-------	--

5. Business Questions and Analysis

Business Question 1: Top 10 Customers by Total Account Balance

Business Objective

To identify the bank's highest-value customers based on the combined balances across all of their accounts.

SQL Techniques Used

- INNER JOIN
- SUM()
- GROUP BY
- ORDER BY
- LIMIT

Business Value

Identifying customers with the highest balances enables the bank to target premium banking services, strengthen customer relationships, and improve customer retention strategies.

Business Question 2: Branch with the Highest Total Account Balance

Business Objective

To determine which bank branch holds the highest total customer account balance.

SQL Techniques Used

- INNER JOIN
- SUM()
- GROUP BY
- ORDER BY

Business Value

This analysis helps management identify top-performing branches and understand where the bank's largest customer deposits are concentrated.

Business Question 3: Customers with More Than One Account

Business Objective

To identify customers who own multiple bank accounts.

SQL Techniques Used

- COUNT()
- GROUP BY

- HAVING
- INNER JOIN

Business Value

Customers with multiple accounts often have stronger relationships with the bank. This information can support customer loyalty programmes and cross-selling opportunities.

Business Question 4: Rank Customers by Total Account Balance

Business Objective

To rank customers according to their combined account balances.

SQL Techniques Used

- Common Table Expression (CTE)
- SUM()
- RANK()
- Window Functions

Business Value

Ranking customers assists relationship managers in identifying high-value clients who may qualify for premium banking products.

Business Question 5: Assign a Unique Rank to Customers

Business Objective

To assign a unique ranking to every customer based on total account balance.

SQL Techniques Used

- CTE
- ROW_NUMBER()
- Window Functions

Business Value

Unlike RANK (), ROW_NUMBER () assigns a unique position to every customer, making it useful for reporting and customer segmentation.

Business Question 6: Running Total of Transactions

Business Objective

To calculate the cumulative transaction value over time.

SQL Techniques Used

- Window Functions
- SUM() OVER()
- ORDER BY

Business Value

Running totals help identify transaction trends and monitor cumulative banking activity over time.

Business Question 7: Running Total of Transactions per Customer

Business Objective

To calculate each customer's cumulative transaction value over time.

SQL Techniques Used

- PARTITION BY
- Window Functions
- SUM() OVER()

Business Value

This analysis provides insight into individual customer transaction behavior and spending patterns.

Business Question 8: Customers Holding Multiple Account Types

Business Objective

To identify customers who own more than one type of bank account.

SQL Techniques Used

- CASE

- MAX()
- COUNT(DISTINCT)
- HAVING

Business Value

Customers with multiple banking products are generally more engaged and may represent valuable long-term clients.

Business Question 9: Branch with the Highest Total Transaction Value

Business Objective

To determine which branch processed the highest total transaction value.

SQL Techniques Used

- Multiple INNER JOINS
- SUM()
- GROUP BY

Business Value

The results help management evaluate branch performance and allocate resources more effectively.

Business Question 10: Top Five Customers by Total Transaction Value

Business Objective

To identify the customers who processed the highest transaction values.

SQL Techniques Used

- SUM()
- GROUP BY
- ORDER BY
- LIMIT

Business Value

High transaction activity may indicate customers who would benefit from specialized financial products or dedicated relationship management.

Business Question 11: Month with the Highest Total Transaction Value

Business Objective

To determine which month recorded the highest banking activity.

SQL Techniques Used

- MONTHNAME()
- YEAR()

- SUM()
- GROUP BY

Business Value

Understanding monthly transaction trends assists the bank in forecasting demand and planning staffing and operational resources.

Business Question 12: Loan Distribution by Status

Business Objective

To analyze the number of loans according to their approval status.

SQL Techniques Used

- COUNT()
- GROUP BY

Business Value

This provides management with an overview of loan approvals, pending applications, and rejected applications.

Business Question 13: Total Loan Amount by Loan Type

Business Objective

To calculate the total value of loans issued for each loan category.

SQL Techniques Used

- SUM()
- GROUP BY

Business Value

This analysis identifies which loan products contribute the most to the bank's lending portfolio.

Business Question 14: Customer Distribution by Province

Business Objective

To determine where the bank's customers are geographically located.

SQL Techniques Used

- COUNT()
- GROUP BY

Business Value

Geographical analysis supports marketing strategies and future branch expansion decisions.

Business Question 15: Account Distribution by Status

Business Objective

To determine how many accounts are currently active or closed.

SQL Techniques Used

- COUNT()
- GROUP BY

Business Value

Monitoring account status provides insight into customer retention and overall account activity.

6. SQL Skills Demonstrated

Throughout this project, a variety of SQL concepts and techniques were applied to solve realistic banking business problems. These skills demonstrate proficiency in relational database design, data manipulation, data aggregation, and analytical reporting.

SQL Skill	Description
Database Design	Designed a relational banking database using multiple related tables.
Primary Keys	Created unique identifiers for each table.
Foreign Keys	Established relationships between related tables.
SELECT Statements	Retrieved data from multiple tables.
INNER JOIN	Combined data across related tables for analysis.
Aggregate Functions	Applied SUM (), COUNT (), and other aggregate functions to summarize data.
GROUP BY	Grouped records to generate business summaries.
HAVING	Filtered grouped data based on aggregate conditions.
ORDER BY	Sorted query results for reporting purposes.
LIMIT	Returned only the most relevant records.

CASE Statements	Categorized and transformed data into business-friendly outputs.
COUNT(DISTINCT)	Counted unique account types for customer analysis.
Common Table Expressions (CTEs)	Improved readability and simplified complex queries.
Window Functions	Performed advanced analytical calculations.
RANK()	Ranked customers based on account balances.
ROW_NUMBER()	Assigned unique rankings to customers.
PARTITION BY	Calculated customer-specific running totals.
Running Totals	Analyzed cumulative transaction values over time.
Business Analysis	Answered real-world banking questions using SQL.

7. Key Findings

The SQL analysis produced several valuable business insights from the banking database.

Customer Analysis

- High-value customers were successfully identified based on their total account balances.
- Several customers hold multiple accounts, demonstrating stronger engagement with the bank.
- Customers with multiple account types present opportunities for additional relationship management and targeted financial services.

Branch Performance

- Significant differences were observed in the total balances and transaction values processed by different branches.

- Identifying the highest-performing branches enables management to investigate successful operational strategies that can be replicated across the organization.

Transaction Analysis

- Transaction trends varied across different months, allowing periods of higher banking activity to be identified.
- Running totals provided a clear view of cumulative transaction growth over time and for individual customers.

Loan Analysis

- Loan information showed a mixture of approved, pending and rejected applications.
- Different loan products contributed varying amounts to the overall lending portfolio.

Customer Distribution

- Customers were distributed across multiple South African provinces.
- Understanding geographic distribution can support future expansion and targeted marketing initiatives.

8. Recommendations

Based on the analysis performed, the following recommendations are proposed.

Recommendation 1: Strengthen Customer Retention

High-value customers identified through account balances should be prioritized for premium banking services and personalized relationship management to improve long-term customer retention.

Recommendation 2: Encourage Product Cross-Selling

Customers who currently own only one account type should be targeted with personalized marketing campaigns encouraging them to open additional banking products such as savings, cheque or investment accounts.

Recommendation 3: Learn from High-Performing Branches

Management should investigate operational practices used by branches with the highest transaction values and customer balances, with the aim of applying successful strategies across other branches.

Recommendation 4: Monitor Transaction Trends

Monthly transaction reporting should be reviewed regularly to improve staffing decisions, operational planning and customer service during peak banking periods.

Recommendation 5: Enhance Database Design

During the project, it was identified that the database does not record which employee opened a customer's account. Adding an EmployeeID field to the **Accounts** table would enable employee performance reporting, improve accountability, and support more detailed operational analysis.

Recommendation 6: Expand Future Analysis

Future versions of the database could include additional information such as customer income, credit scores, loan repayment history and digital banking activity. These enhancements would enable more advanced analytics and predictive modelling.

9. Conclusion

The Banking Database Analysis Project successfully demonstrated the practical application of SQL within a simulated banking environment. A relational database was designed, populated with realistic sample data and analyzed using a variety of SQL techniques to answer business-focused questions.

The project provided practical experience in database design, table relationships, SQL querying, data aggregation, window functions and analytical reporting. Beyond technical implementation, the project emphasized the importance of interpreting data to generate meaningful business insights and recommendations.

Overall, this project strengthened skills in relational database management and business analysis while demonstrating the ability to solve real-world data problems using SQL. The knowledge and techniques applied throughout the project provide a strong foundation for further development in data analytics, business intelligence and data visualization.